SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I Sanjana.G(192511020) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Sanjana.G

Assessment Tasks

Concept Mapping Tasks

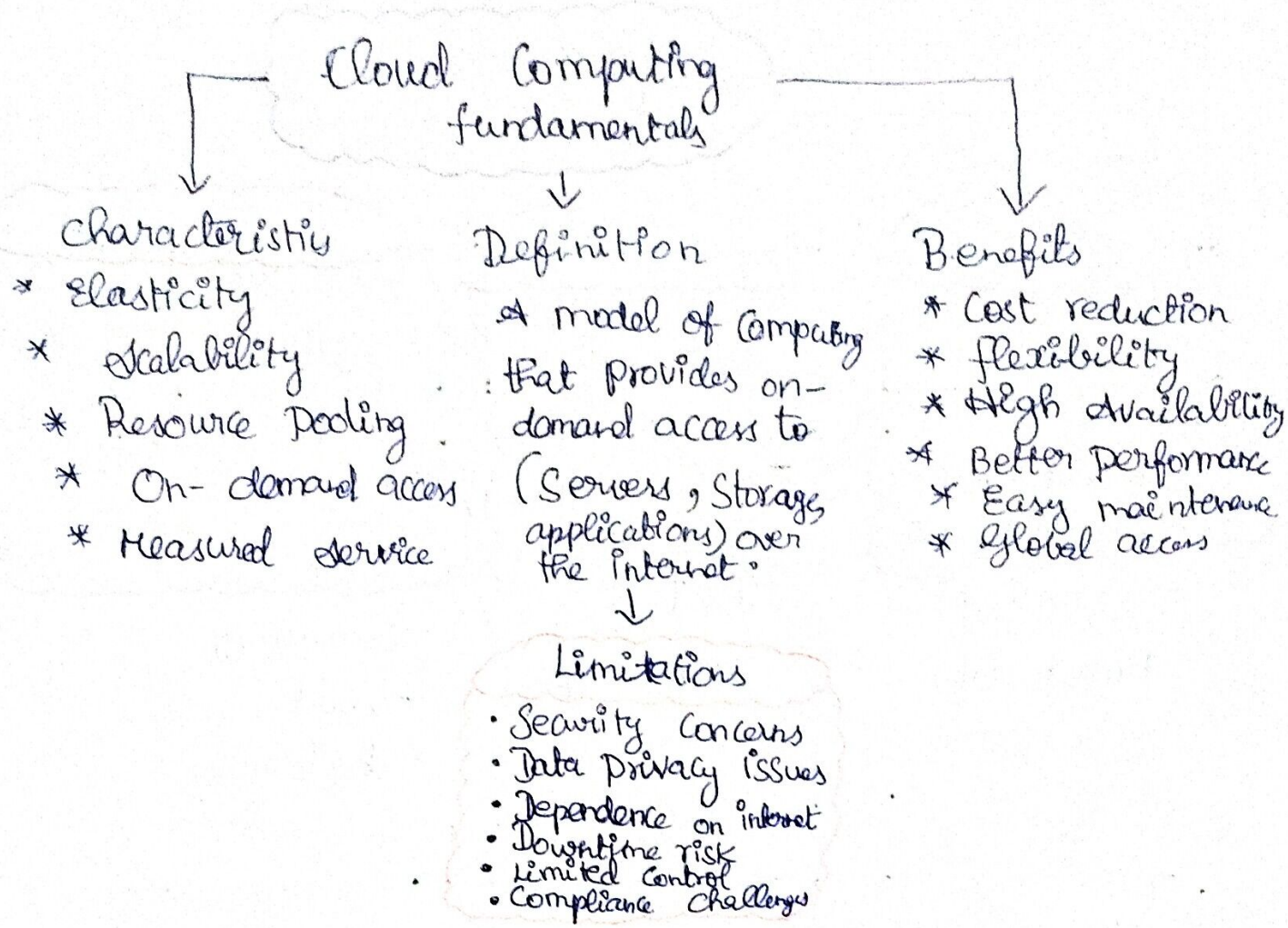
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

1) Concept Map 1:



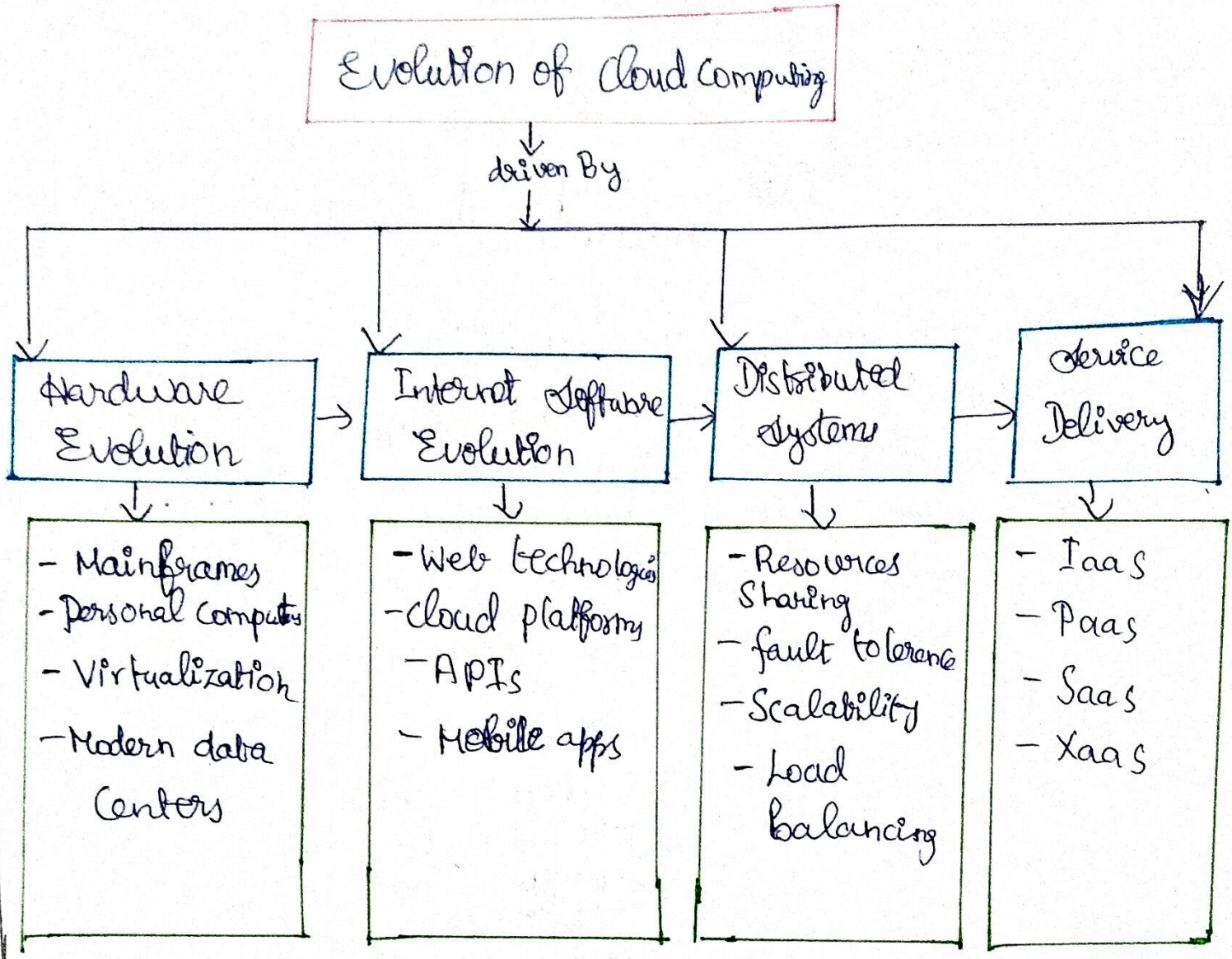
Explanation:

Cloud computing is a model that provides on-demand access to shared resources like servers, storage and applications through the internet.

The main characteristics are Elasticity, Scalability, resource Pooling, On-demand access and measured service.

Cloud computing provides benefits such as cost reduction, flexibility, high availability, better performance and global access. It also have limitation like security concerns, data privacy issues and limited control.

2) Concept Map 2:

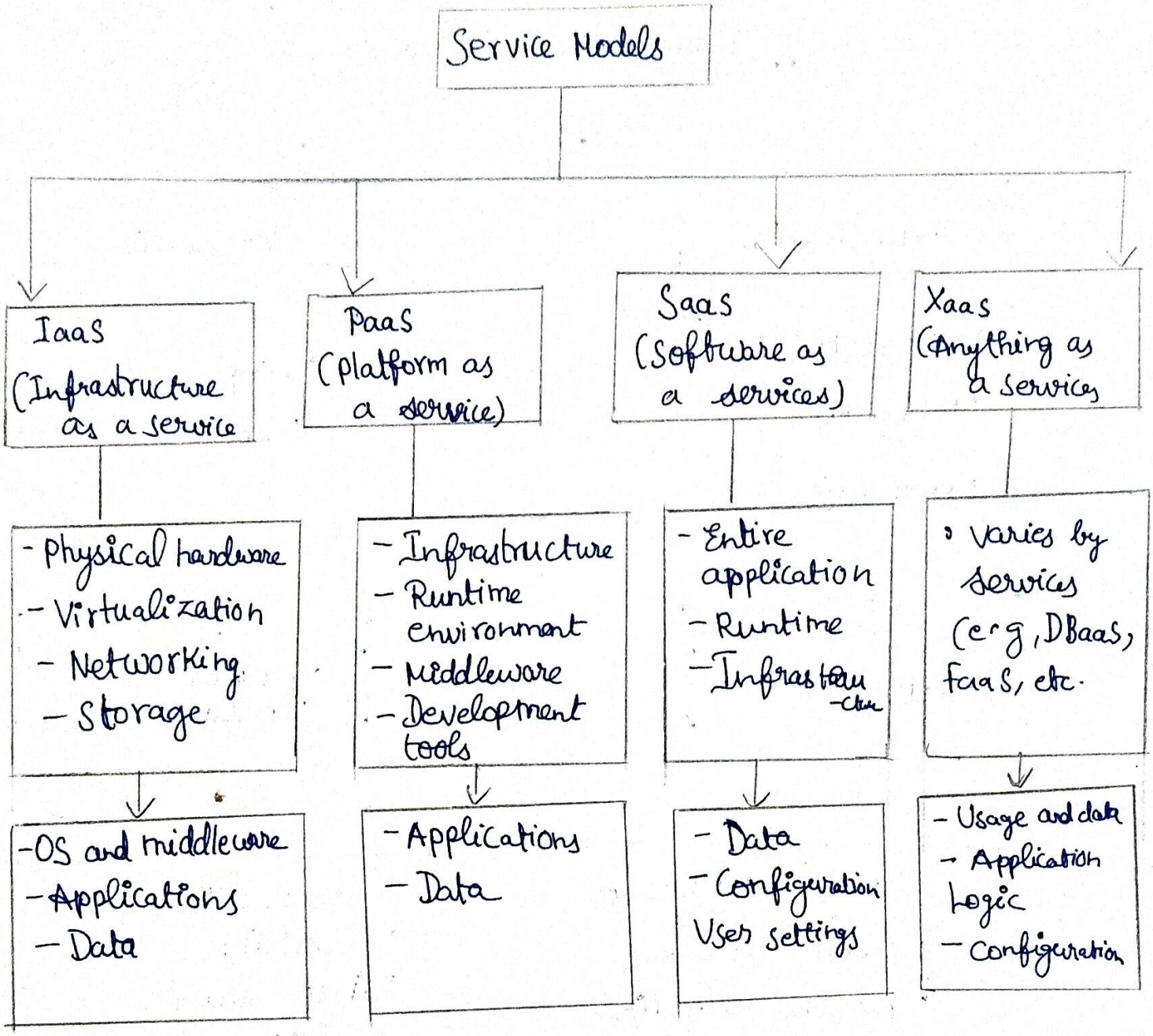


Explaination:

Cloud computing evolved from physical servers and mainframes to virtualization and modern data centres. Internet software developed from static websites to dynamic web applications and cloud-Based applications.

Distributed system allows multiple computer to work together led to cloud service models such as Iaas, Paas, Saas and Xaas

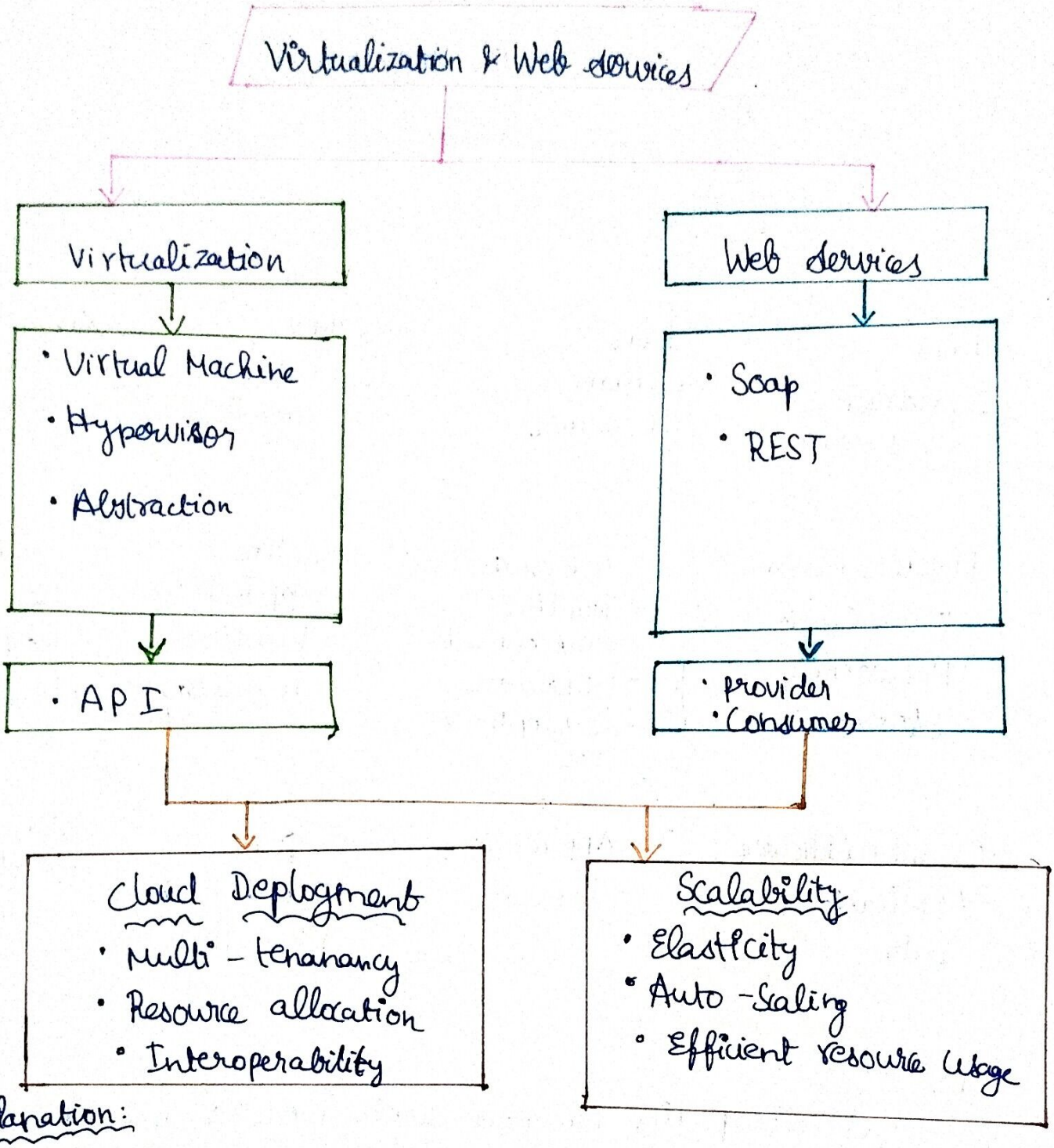
3) Concept Map: 3 Service Models



Explanation:

Cloud computing has four service models: Iaas, Paas, Saas and Xaas. Iaas provides Virtual machines, storage, and networking. Paas provides Platforms and tools for application development. Saas offers ready-to-use software through the internet. Xaas includes various services such as databases, functions, and containers. The responsibilities of providers and users differ depending on the service model.

1) Concept Map:1



Explanation:

Virtualization Creates multiple virtual machines on one server using a hypervisor, reducing costs and improving resource use. Web services like SOAP and Rest enable communication between system. Together, they support cloud deployment, flexibility and scalability.

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented